

DASR Supplement 1: Detailed Empirical Results

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1 Scope and reading rules

This document expands the empirical record of the paper without redefining the tasks or operating settings. The shared task specifications are maintained separately. Each row below is one task crossed with one evidence-cost schedule and one action-cost profile. The nine rows per task reuse the same sealed-audit subjects and are operating sensitivities, not independent replications. Negative DASR-minus-comparator differences favor DASR. Paired intervals are descriptive 95% intervals on identical sealed-audit cases.

Final DASR is one frozen frontier containing score anchors, independently constructed path-free policies, and path-bearing policies. Policy Selection chooses the least-cost candidate passing the pointwise 0.045 screen. Exactly that one policy per cell accesses the Sealed Audit split at 0.05.

DASR generalization to new task family and structure resolution based on multiple model architectures remains future work.

2 Comparator standings

Comparator	Point W/T/L	Interval F/T/U/A	Mean difference	Mean route disagreement
Calibrated-score representation	26/1/0	26/1/0/0	-0.398365	0.155689
Fast Yet Safe	26/0/1	26/0/0/1	-0.728210	0.308175
MSRC	27/0/0	27/0/0/0	-4.457654	0.719465
Ringel accumulated-gap	27/0/0	26/0/1/0	-2.654081	0.666896
Wang sequential LP	27/0/0	27/0/0/0	-4.472070	0.730353

F/T/U/A means interval favors DASR, exact tie, unresolved, or favors the comparator. Ringel’s one unresolved comparison is MDS, Exp. 1.5, Balanced: difference -0.033887 , interval $[-0.070889, 0.003115]$.

3 Complete 27-cell comparison

All 27 DASR, calibrated-score, and Fast Yet Safe selections pass the sealed tail audit. “Path” denotes the path-bearing information class; “path-free” denotes the independent path-free construction branch.

Task	Evidence	Action	Branch	DASR	Cal. score	Fast	Δ score	Δ Fast	Tail / Review
CFPB	Exp. 1.333	Balanced	path	13.0044	13.2298	14.2222	-0.2254	-1.2178	0.0361 / 21.8%
CFPB	Exp. 1.333	FP strict	path-free	13.4285	13.6856	15.1302	-0.2571	-1.7017	0.0387 / 62.7%
CFPB	Exp. 1.333	Higher downstream	path-free	14.7917	15.4256	16.1523	-0.6339	-1.3606	0.0491 / 20.0%
CFPB	Exp. 1.5	Balanced	path	12.5706	12.8204	13.9606	-0.2498	-1.3899	0.0361 / 21.8%
CFPB	Exp. 1.5	FP strict	path-free	13.1886	13.4297	14.8686	-0.2411	-1.6800	0.0387 / 62.7%
CFPB	Exp. 1.5	Higher downstream	path-free	14.4069	14.8142	15.8907	-0.4073	-1.4838	0.0421 / 21.7%
CFPB	Linear	Balanced	path-free	13.6034	13.8472	14.8538	-0.2438	-1.2504	0.0361 / 54.9%
CFPB	Linear	FP strict	path-free	14.0000	14.0000	15.7618	-0.0000	-1.7618	0.0000 / 100.0%
CFPB	Linear	Higher downstream	path-free	15.7764	16.2480	16.7839	-0.4716	-1.0075	0.0480 / 20.0%
INSPIRE	Exp. 1.333	Balanced	path-free	6.8846	7.4889	7.4888	-0.6043	-0.6042	0.0476 / 29.1%
INSPIRE	Exp. 1.333	FP strict	path-free	6.9060	7.4889	7.5217	-0.5830	-0.6157	0.0476 / 29.7%
INSPIRE	Exp. 1.333	Higher downstream	path-free	7.9293	8.6776	8.3845	-0.7483	-0.4551	0.0461 / 24.7%
INSPIRE	Exp. 1.5	Balanced	path-free	6.5830	7.1726	7.2880	-0.5896	-0.7050	0.0476 / 29.1%
INSPIRE	Exp. 1.5	FP strict	path-free	6.6044	7.1726	7.3209	-0.5682	-0.7165	0.0476 / 29.7%
INSPIRE	Exp. 1.5	Higher downstream	path-free	7.6759	8.3547	8.1837	-0.6788	-0.5077	0.0461 / 24.7%
INSPIRE	Linear	Balanced	path-free	7.5944	8.2712	8.0340	-0.6768	-0.4396	0.0461 / 24.7%
INSPIRE	Linear	FP strict	path-free	7.6162	8.2712	8.0669	-0.6550	-0.4507	0.0461 / 25.4%
INSPIRE	Linear	Higher downstream	path-free	8.9707	9.3595	8.9297	-0.3889	+0.0410	0.0491 / 19.5%
MDS	Exp. 1.333	Balanced	path	7.4402	7.6623	7.5996	-0.2221	-0.1595	0.0362 / 36.3%
MDS	Exp. 1.333	FP strict	path	7.5141	7.6441	7.6642	-0.1300	-0.1501	0.0362 / 38.5%
MDS	Exp. 1.333	Higher downstream	path	8.8020	9.2675	9.2648	-0.4655	-0.4628	0.0414 / 30.3%
MDS	Exp. 1.5	Balanced	path	7.2598	7.4814	7.4587	-0.2215	-0.1988	0.0362 / 36.3%
MDS	Exp. 1.5	FP strict	path	7.2405	7.4631	7.5232	-0.2226	-0.2827	0.0421 / 34.3%
MDS	Exp. 1.5	Higher downstream	path	8.5197	8.9925	9.1292	-0.4728	-0.6095	0.0411 / 31.1%
MDS	Linear	Balanced	path	7.8989	8.1216	7.9653	-0.2227	-0.0664	0.0362 / 36.3%
MDS	Linear	FP strict	path	7.9728	8.1034	8.0299	-0.1306	-0.0570	0.0362 / 38.5%
MDS	Linear	Higher downstream	path	9.2630	9.7083	9.6305	-0.4453	-0.3675	0.0413 / 32.1%

The one exact calibrated-score tie is CFPB with linear evidence prices and FP-strict action costs: both choose full expert review. The sole Fast Yet Safe win is INSPIRE with linear evidence prices and higher downstream-action cost: DASR minus Fast is +0.040994 with paired interval [0.001460, 0.080529].

4 Representation and component evidence

Representation	Audit passes	Workload passes	CFPB	INSPIRE	MDS
DASR	27/27	26/27	13.8634	7.4183	7.9901
Calibrated-score representation	27/27	26/27	14.1667	8.0286	8.2716
Dense prefix raw	27/27	4/27	14.3444	15.3333	15.1605
Dense prefix HS	27/27	8/27	14.0703	15.3333	15.1605
Dense path HS	27/27	8/27	14.1900	15.3333	15.2124
Expanded path-HS sensitivity	27/27	16/27	14.1063	9.0509	15.2124

Hard audit success alone does not imply useful routing: dense clinical representations often satisfy the boundary by routing nearly every case to review. DASR is cheaper in all 16 settings where expanded path HS and DASR both pass the soft workload diagnostic.

The two final component restrictions are nested policy-frontier comparisons. Removing calibrated regions yields DASR wins/ties/losses of 26/1/0 and mean difference -0.387087 . Removing the path family yields 9/16/2 by report point cost: all nine MDS settings favor paths, all nine INSPIRE settings tie, and the two CFPB reversals are small and interval-unresolved. On Policy Selection data, adding paths improves 11/27 cells and cannot worsen the least-cost survivor by construction.

5 Limited construction outcomes

The reduced-outcome study uses the registered central contract for each task and five subject-level samples at 50% and 75% construction budgets. Each structural and score-only policy is rebuilt from the same sampled construction units. Every selected policy passes Sealed Audit.

Task	Budget	DASR	Cal. score	Mean difference	Structural / 5
CFPB	50%	13.0155	13.2309	-0.2154	5
CFPB	75%	13.0448	13.2298	-0.1850	5
INSPIRE	50%	7.0341	7.5506	-0.5166	5
INSPIRE	75%	6.9458	7.5173	-0.5715	5
MDS	50%	7.4209	8.9635	-1.5425	5
MDS	75%	7.4218	7.6815	-0.2598	5

All 30 structural policies are cheaper than the matched calibrated-score representation; 29 paired intervals favor structure and one CFPB 75% interval is unresolved. The MDS 50% mean is inflated by one calibrated-score full-review fallback; its five-seed median saving is 0.227413.

6 Solver sensitivity

The following central-contract comparison freezes the representation and changes the optimizer.

Task	LP + score	CMDP + score	LP + DASR	CMDP + DASR
CFPB	13.2298	13.2522	13.0044	13.0615
INSPIRE	7.4889	7.6227	7.2957	6.8107
MDS	7.6623	14.0000	7.4402	13.8196

Structure improves upon score states under both solvers in all three tasks. Iterative dual ascent routes 96.56% of MDS cases to review, motivating the occupation-measure LP as the stable deployment optimizer.

7 Frozen-frontier comparison diagnostics

We replayed the frozen frontiers on the registered Policy Selection batches without new construction or audit access. The disagreement-localized paired radius was tighter than the global-range radius in every setting, with a mean ratio of 0.256. These quantities are comparison diagnostics under the registered executor; the safety statement comes from the independent audit of the selected policy.

8 Operator evidence

All 27 selected policies are covered by 14 exact behavior-hash reuses and 13 deterministic registered-batch selected-policy replays. Across 52 applicable direct edits, 40 pass screen and audit, 12 stop before Sealed Audit access, and the maximum off-address route-change count is zero. The central MDS address changes 162 intended routes and no others; the nearest scalar threshold changes seven additional routes.

For each selected table, its five-coordinate consequence card reproduces report cost at all nine vertices to maximum error 1.776×10^{-15} . The saved eligible libraries contain seven, five, and six unique routes for CFPB, INSPIRE, and MDS; their minimax maximum vertex regrets are 1.089632, 0.082075, and 0.081359.

9 Selected-policy execution semantics

The selected-policy execution audit replays all 27 selected policies without a new LP solve, selector change, or unselected Sealed Audit access. Eight policies have pure occupied state-action rows. Nineteen contain a mixed row and exhibit nonzero partition dependence under deterministic halves, quarters, and 20 fixed-seed subject-preserving partitions. The maximum route disagreement is 0.005069, maximum absolute cost drift is 0.006596, and maximum absolute tail-UCB drift is 0.004883. Every full registered-batch replay exactly matches the stored selected decisions.

Accordingly, the current executor is deterministic for a registered batch: it realizes stored action fractions by calibrated-score ordering, deterministic integer allocation, and stable tie-breaking within each reached state. Pure rows are casewise actions; mixed rows are cohort-relative allocations. The reported costs and audits are unchanged and exact for their registered evaluation batches.